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BSIT 301I

## **Week 10 Lab Activity: Integrating Applications**

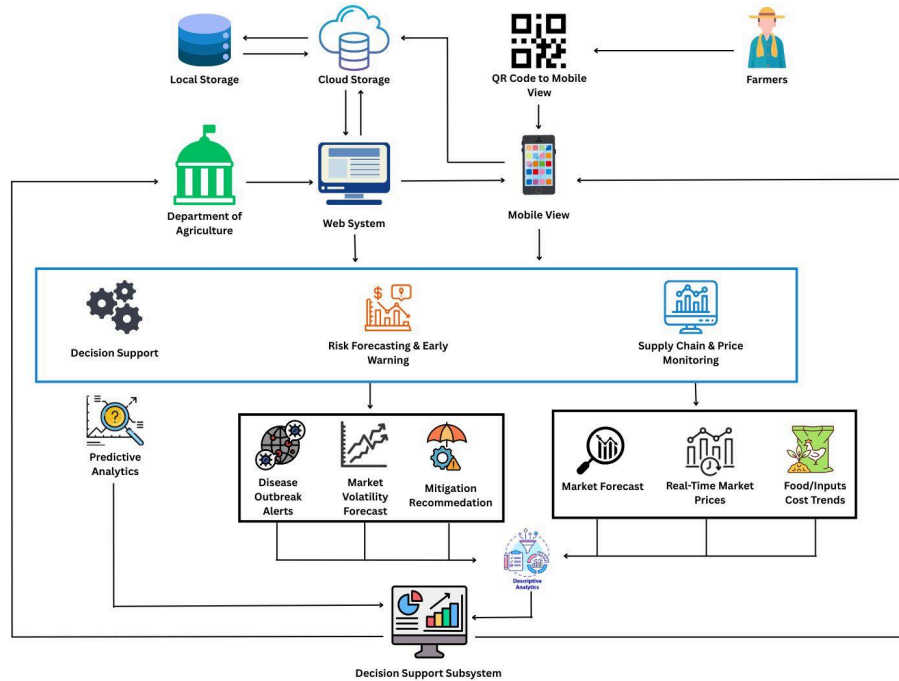
### **1. Identify Sub-Systems**

- \* Supply Chain and Price Monitoring Sub-System
- \* Risk Forecasting and Early Warning Sub-System
- \* Decision Support Dashboard Sub-System

### **2. Define Features of Sub-Systems**

- \* Supply Chain and Price Monitoring Sub-System
  - Farmgate Price Collection
  - Retail Price Integration
  - Production Data Management
  - Historical Data Repository
  - Data Validation and Cleaning
  - Price Comparison Engine
- \* Risk Forecasting and Early Warning Sub-System
  - Price Forecasting
  - Optimal Selling Time Prediction
  - Supply Shortage Forecasting
  - Disease Outbreak Early Warning
  - Seasonal Risk Assessment
  - Calamity Impact Analysis
- \* Decision Support Dashboard Sub-System
  - Mobile:
    - Real-Time Price Comparison
    - Personalized Alerts and Notifications
    - Sell Timing Recommendations
    - Offline Mode
  - Web:
    - Regional Price Trend Monitoring
    - Farmgate-Retail Gap Analysis
    - Production Forecast Visualization
    - Risk Heat Maps
    - Automated Report Generation

### 3. Create an Input-Process-Output Diagram



A comprehensive, data-driven agricultural Decision Support Subsystem aimed to close the gap between field-level operations and sophisticated analytics is depicted in the conceptual framework. The system uses secure cloud and local storage to feed a centralized web hub with real-time inputs from farmers via mobile applications and institutional data from the Department of Agriculture. Fundamentally, the framework uses both descriptive and predictive analytics to produce important insights about risk forecasting, including market volatility and disease outbreak alerts, as well as thorough supply chain and price monitoring. In the end, these intricate data streams are combined into a centralized subsystem, generating an ongoing feedback loop that sends market intelligence and practical mitigation suggestions straight to farmers' and policymakers' mobile devices, enabling them to make well-informed agricultural decisions.